

linux end to end cheat sheet

Whoami:displays the current user account

Ls: list of files and directory .ipconfig:lists your ip address

```
/dev/disk7s2 2.1Gi 2.0Gi 170Mi 93%
[waseemakram@waseem ~ % whoami
waseemakram
[waseemakram@waseem ~ % ls
7095328761.pub      bash      devops      Downloads
alphabet           david     dir1        farhan
Applications       Desktop  Documents   first
[waseemakram@waseem ~ % ip a
zsh: command not found: ip
[waseemakram@waseem ~ % ip config
zsh: command not found: ip
[waseemakram@waseem ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
```

Passwd: used to change the password of user . Touch used for creating a 0kb file

touch :~ used for creating a empty file

```
[waseemakram@waseem ~ % passwd
Changing password for waseemakram.
Old Password:
[waseemakram@waseem ~ % touch f1
[waseemakram@waseem ~ % pwd
/Users/waseemakram
[waseemakram@waseem ~ % man ps -ef

zsh: suspended  man ps -ef
[waseemakram@waseem ~ % gprags "f1"
```

find command for finding the file

```
waseemakram@waseem ~ % find . -name "*.txt"
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/licenses/LICENSE.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/licenses/AUTHORS.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/licenses/src/pip/_vendor/dependency_groups/LICENSE.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/licenses/src/pip/_vendor/distlib/LICENSE.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/licenses/src/pip/_vendor/cachecontrol/LICENSE.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/licenses/src/pip/_vendor/urllib3/LICENSE.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/entry_points.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip-25.2.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/crcmod-1.7.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/certifi-2025.10.5.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/cffi-2.0.0.dist-info/entry_points.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/cffi-2.0.0.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/cryptography-42.0.7.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/pip/_vendor/vendor.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/google_crc32c-1.7.1.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/setuptools-80.9.0.dist-info/entry_points.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/setuptools-80.9.0.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/setuptools/_vendor/jaraco.text-3.12.1.dist-info/top_level.txt
./config/gcloud/virtenv/lib/python3.12/site-packages/setuptools/_vendor/jaraco.collections-5.1.0.dist-info/top_level.txt
```

Essential File Management Tools Commands

Ls:Lists files and directories within a specified path.This command is used to view the contents of Description

```
waseemakram@waseem waseem % ls /Users/waseemakram/waseem
akram          akram.git      app            games          horizon
waseemakram@waseem waseem % cd
```

Man : for command details. apropos: for command related to a specific task on disrcp

```
waseemakram@waseem ~ % apropos ls
git-credential(1)      - Retrieve and store user credentials
git-credential-cache--daemon(1) - Temporarily store user credentials in memory
git-credential-store(1) - Helper to store credentials on disk
git-difftool(1)        - Show changes using common diff tools
git-ls-files(1)        - Show information about files in the index and the working tree
git-ls-remote(1)       - List references in a remote repository
git-ls-tree(1)         - List the contents of a tree object
git-mailsplit(1)       - Simple UNIX mbox splitter program
git-mergetool(1)       - Run merge conflict resolution tools to resolve merge conflicts
git-mktree(1)          - Build a tree-object from ls-tree formatted text
git-send-email(1)      - Send a collection of patches as emails
gitcredentials(7)      - Providing usernames and passwords to Git
gitprotocol-common(5)  - Things common to various protocols
gitprotocol-http(5)    - Git HTTP-based protocols
nmedit(1)              - change global symbols to local symbols
strip(1)               - remove symbols
unifdef(1), unifdefall(1) - remove preprocessor conditionals from code
xcrun(1)               - Run or locate development tools and properties
AllPlanes(3), BlackPixel(3), WhitePixel(3), ConnectionNumber(3), DefaultColormap(3), DefaultDepth(3), XExtendedMaxRequestSize(3), LastKnownRequestProcessed(3), NextRequest(3), ProtocolVersion
BI0_ctrl(3ssl), BI0_callback_ctrl(3ssl), BI0_ptr_ctrl(3ssl), BI0_int_ctrl(3ssl), BI0_reset(3ssl)
g(3ssl), BI0_get_info_callback(3ssl), BI0_set_info_callback(3ssl), BI0_info_cb(3ssl), BI0_get_kt
BN_set_bit(3ssl), BN_clear_bit(3ssl), BN_is_bit_set(3ssl), BN_mask_bits(3ssl), BN_lshift(3ssl),
```

Cp: Copies files from one location to another

Mkdir: for creating directory

Rm :for removing a file

Tar:archives multiple files into single file

Ex: tar -cvf archive.tar/folder creates a archive of the folder

tar -xvf is used for extracting the tar file

```
waseemakram@waseem waseem % tar -cvf horizon.tar horizon
a horizon
waseemakram@waseem waseem % ls
akram          akram.git      app            file1          games          horizon        horizon.tar
waseemakram@waseem waseem % cd horizon.tar
cd: not a directory: horizon.tar
waseemakram@waseem waseem % cat horizon.tar
horizon000755 000765 000024 000000002000 15101125417 014121 0ustar00waseemakramstaff0000000 000000
waseemakram@waseem waseem % tar xvf horizon.tar
x horizon
```

```

hello world and horizon
[waseemakram@waseem waseem % touch file1
[waseemakram@waseem waseem % vi file2
[waseemakram@waseem waseem % cp file2 file1
[waseemakram@waseem waseem % cat file1
hello copying the file
[waseemakram@waseem waseem % mkdir folder1
[waseemakram@waseem waseem % ls
akram          akram.git      app          file1        file2        folder1      games        horizon
[waseemakram@waseem waseem % rmdir folder1
[waseemakram@waseem waseem % ls
akram          akram.git      app          file1        file2        games        horizon
[waseemakram@waseem waseem % rm file2
[waseemakram@waseem waseem % ls
akram          akram.git      app          file1        games        horizon
[waseemakram@waseem waseem % █

```

cp :~ is used for copying the file from one location to another file

Gzip file1 compresses file1 which reduces size

```

akram          akram.git      app          file1        games
[waseemakram@waseem waseem % gzip file1
[waseemakram@waseem waseem % ls -ltr
total 24
-rw-r--r--  1 waseemakram  staff    0 22 Oct 14:55 akram.git
-rw-r--r--  1 waseemakram  staff    0 22 Oct 16:10 app
-rw-r--r--  1 waseemakram  staff    0 22 Oct 16:28 games
-rw-r--r--  1 waseemakram  staff   49 31 Oct 17:11 file1.gz
lrwxr-xr-x  1 waseemakram  staff    5 31 Oct 17:18 soft.lnk -> file1

```

Bzip2: a file compression utility it offers better compress ratio than gzip

Xz :command uses for high compress utility provide best compress ratio

File:useful for identifying file formats

```

dr.: /waseem/: No such file or directory
[waseemakram@waseem waseem % bzip2 file1
[waseemakram@waseem waseem % ls
akram          akram.git      app          file1.bz2    file1.gz     games        horizon      horizon.tar  soft.lnk
[waseemakram@waseem waseem % cat file1.bz2
[BZh91AY&SY}3NQ?@&f?? "???d??P?y?*be100o_rE8P?}3N?
[waseemakram@waseem waseem % xz file1
xz: file1: No such file or directory
[waseemakram@waseem waseem % xz file1.gz
[waseemakram@waseem waseem % ls
akram          akram.git      app          file1.bz2    file1.gz.xz  games        horizon      horizon.tar  soft.lnk
[waseemakram@waseem waseem % cat file1.gz.xz
[??zXZ?F?51!?!?i?file1?H???WH?/???KW(?HUH??I?4K00Q1?????}Y
[waseemakram@waseem waseem % file games
games: empty
[waseemakram@waseem waseem % file soft.lnk
soft.lnk: broken symbolic link to file1
[waseemakram@waseem waseem % █

```

Zip: command is used for packing and compressing archive file

gunzip :~ unzips a file

```

soft.lnk: broken symbolic link to file1
[waseemakram@waseem waseem % touch file4
[waseemakram@waseem waseem % vi file2
[waseemakram@waseem waseem % zip file4.zip file4
adding: file4 (stored 0%)
[waseemakram@waseem waseem % ls
akram          akram.git      app          file1.bz2    file1.gz.xz  file2        file4        file4.zip
[waseemakram@waseem waseem % unzip -l file4.zip
Archive:  file4.zip
Length  Date    Time    Name
-----  -
0       11-02-2025 12:39  file4
-----  -
0       11-02-2025 12:39  1 file

```


Df -h :for human readability in mb and gb not in big numbers

```
waseemakram@waseem:~$ df -h
```

Filesystem	Size	Used	Avail	Capacity	iused	ifree	%iused	Mounted on
/dev/disk3s1s1	228Gi	10Gi	120Gi	9%	426k	1.3G	0%	/
devfs	198Ki	198Ki	0Bi	100%	684	0	100%	/dev
/dev/disk3s6	228Gi	2.0Gi	120Gi	2%	2	1.3G	0%	/System/Volumes/VM
/dev/disk3s2	228Gi	6.6Gi	120Gi	6%	1.2k	1.3G	0%	/System/Volumes/Preboot
/dev/disk3s4	228Gi	4.1Mi	120Gi	1%	54	1.3G	0%	/System/Volumes/Update
/dev/disk1s2	500Mi	6.0Mi	483Mi	2%	1	4.9M	0%	/System/Volumes/xarts
/dev/disk1s1	500Mi	5.4Mi	483Mi	2%	33	4.9M	0%	/System/Volumes/iSCPreboot
/dev/disk1s3	500Mi	928Ki	483Mi	1%	85	4.9M	0%	/System/Volumes/Hardware
/dev/disk3s5	228Gi	88Gi	120Gi	43%	928k	1.3G	0%	/System/Volumes/Data
map auto_home	0Bi	0Bi	0Bi	100%	0	0	-	/System/Volumes/Data/home
/Users/waseemakram/Downloads/Visual Studio Code.app	228Gi	88Gi	122Gi	43%	925k	1.3G	0%	/private/var/folders/g7/ldjy8qpd1458ss3521
/Users/waseemakram/Downloads/Sublime Text.app	228Gi	88Gi	122Gi	43%	925k	1.3G	0%	/private/var/folders/g7/ldjy8qpd1458ss3521

Working with Text Files Commands

Less: similar to more, but provides more control over viewing text.file

```
[waseemakram@waseem ~ % vim tutor
[waseemakram@waseem ~ % vi david
[waseemakram@waseem ~ % vim david
[waseemakram@waseem ~ % more david
Akram ,21 ,Hyderabad ,50000
Farhan ,25 ,Chennai ,60000
name ,age ,city ,salary
Waseem ,23 ,Delhi ,55000
[waseemakram@waseem ~ % less david
```

Vimtutor :a tutorial for learning vim an advanced editor


```

= Welcome to the VIM Tutor - Version 1.7 =
= CHAPTER ONE =
=

Vim is a very powerful editor that has many commands, too many to
explain in a tutor such as this. This tutor is designed to describe
enough of the commands that you will be able to easily use Vim as
an all-purpose editor.
The approximate time required to complete the tutor is 30 minutes,
depending upon how much time is spent with experimentation.

ATTENTION:
The commands in the lessons will modify the text. Make a copy of this
file to practice on (if you started "vimtutor" this is already a copy).

It is important to remember that this tutor is set up to teach by
use. That means that you need to execute the commands to learn them
properly. If you only read the text, you will forget the commands!
Now, make sure that your Caps-Lock key is NOT depressed and press
the j key enough times to move the cursor so that lesson 1.1.1
completely fills the screen.

```

Lesson 1.1.1: MOVING THE CURSOR

```

** To move the cursor, press the h,j,k,l keys as indicated. **
      ^
      k      Hint: The h key is at the left and moves left.
    < h  l >      The l key is at the right and moves right.
      j      The j key looks like a down arrow.
      v

```

1. Move the cursor around the screen until you are comfortable.
2. Hold down the down key (j) until it repeats.
Now you know how to move to the next lesson.
3. Using the down key, move to lesson 1.1.2.

NOTE: If you are ever unsure about something you typed, press <ESC> to place
you in Normal mode. Then retype the command you wanted.

NOTE: The cursor keys should also work. But using hjkl you will be able to
move around much faster, once you get used to it. Really!

Lesson 1.1.2: EXITING VIM

!! NOTE: Before executing any of the steps below, read this entire lesson!!

1. Press the <ESC> key (to make sure you are in Normal mode).
2. Type: :q! <ENTER>.
This exits the editor, DISCARDING any changes you have made.
3. Get back here by executing the command that got you into this tutor. That
might be: vimtutor <ENTER>
4. If you have these steps memorized and are confident, execute steps
1 through 3 to exit and re-enter the editor.

NOTE: :q! <ENTER> discards any changes you made. In a few lessons you
will learn how to save the changes to a file.

5. Move the cursor down to lesson 1.1.3.

Lesson 1.1.3: TEXT EDITING - DELETION

```

** Press x to delete the character under the cursor. **
-- INSERT --

```

Head : command will show upper lines of the file

Tail : shows bottom lines of the file

Cat: displays the entire content of the file

```

zsh: suspended less david
[waseemakram@waseem ~ % head david
Akram ,21 ,Hyderabad ,50000
Farhan ,25 ,Chennai ,60000
name ,age ,city ,salary
Waseem ,23 ,Delhi ,55000
[waseemakram@waseem ~ % head -1 david
Akram ,21 ,Hyderabad ,50000
[waseemakram@waseem ~ % tail -1 david
Waseem ,23 ,Delhi ,55000
[waseemakram@waseem ~ % cat david
Akram ,21 ,Hyderabad ,50000
Farhan ,25 ,Chennai ,60000
name ,age ,city ,salary
Waseem ,23 ,Delhi ,55000
[waseemakram@waseem ~ %

```

Advanced Text File Processing Commands

awk: additional feature for text manipulation

Sed: modifying text directly from command line

Tr: translates or deletes character from input text

Sort: sort text files in alphabetical order

```
waseemakram@waseem ~ % awk '{print $1}' david
Akram
Farhan
name
Waseem
waseemakram@waseem ~ % sed 's/Farhan/irfan/' david
Akram ,21 ,Hyderabad ,50000
irfan ,25 ,Chennai ,60000
name ,age ,city ,salary
Waseem ,23 ,Delhi ,55000
waseemakram@waseem ~ % tr '[:lower:]' '[:upper:]' < david
AKRAM ,21 ,HYDERABAD ,50000
FARHAN ,25 ,CHENNAI ,60000
NAME ,AGE ,CITY ,SALARY
WASEEM ,23 ,DELHI ,55000
waseemakram@waseem ~ % sort david
Akram ,21 ,Hyderabad ,50000
Farhan ,25 ,Chennai ,60000
name ,age ,city ,salary
Waseem ,23 ,Delhi ,55000
waseemakram@waseem ~ %
```

Connecting to a Server Commands

Ssh : stands for secure shell it is used to connect and log in to a remote

```
waseemakramewaseem downloads % ssh -i waseem.pem ec2-user@13.50.240.232

A newer release of "Amazon Linux" is available.
Version 2023.9.20251020:
Version 2023.9.20251027:
Run "/usr/bin/dnf check-release-update" for full release and version update info

,      #_
~\    #####_      Amazon Linux 2023
~~~  \#####\
~~~~ \###|
~~~~  \#/_  ____  https://aws.amazon.com/linux/amazon-linux-2023
~~~~   V~'  '->
~~~~
~~~~_. _ _/_
~~~~/_/_/_
~~~~/_m/'

Last login: Thu Oct 30 11:44:21 2025 from 13.48.4.203
[ec2-user@ip-172-31-7-102 ~]$ ls
```

Scp : used to secure copy from one machine to another

```

waseemakram@waseem ~ % scp -i ~/Downloads/waseem.pem ec2-user@13.50.105.48:/home/ec2-user/file1 ~/Downloads/project-folder/
file1
waseemakram@waseem ~ % ls
01-Devops_Introduction.mp4
305290863_769247980843493_2396714656254633878_n_597852462321174.jpg
5160388
IMG_20220516_134649 (1).jpg
IMG_20220516_134649.jpg
peakpx.jpg
project-folder

```

Working with the Bash Shell commands

History : for used commands history

Alias : create short cut for long commands

```

waseemakram@waseem ~ % history -d 1171
1171 16:47 tee
1172 16:47 ls | tee david
1173 16:48 ls
1174 16:48 history
waseemakram@waseem ~ % history
1160 awk '{print $1}' david
1161 awk 's/Farhan/irfan' david
1162 sed 's/Farhan/irfan' david
1163 "sed 's/Farhan/irfan' david"
1164 "sed 's/Farhan/irfan/' david"
1165 awk '{print $1}' david
1166 sed 's/Farhan/irfan/' david
1167 tr '[:lower:]' '[:upper:]' < david
1168 sort david
1169 su
1170 sudo -i
1171 tee
1172 ls | tee david
1173 ls
1174 history
1175 history -d 1171
waseemakram@waseem ~ % alias ll='ls -i'
waseemakram@waseem ~ % ll
7654661 7095328761.pub          7867478 david                265767 Downloads             7686263 john2                265717 Music
7478077 alphabet                265721 Desktop              7768086 f1                   265723 Library               6328051 mydir
1004905 Applications            7432188 devops              6664782 farhan               7702730 marks.file           6284625 myfolder
7839200 aseem                  7627171 dir1                 4939608 first                 6328396 mohd2                265719 Pictures
7383556 bash                   265765 Documents             7686098 john_tar               265763 Movies                 265759 Public
waseemakram@waseem ~ % export

```

Creating a group and adding member

```

zsh: command not found: usermod
waseemakram@waseem ~ % sudo dscl . -create /groups/developers
[Password:
waseemakram@waseem ~ % sudo dscl . -create /groups/developers PrimaryGroupID 510
waseemakram@waseem ~ % sudo dscl . -create /groups/developers groupmembership waseem
waseemakram@waseem ~ % sudo dscl . -append /groups/developers groupmembership waseem
waseemakram@waseem ~ % sudo dscl . -read /groups/developers
dsAttrTypeNative:groupmembership: waseem waseem
dsAttrTypeNative:record_daemon_version: 9054000
AppleMetaNodeLocation: /Local/Default
GeneratedUID: C37E0F5B-2BF7-4903-BD8F-DCA498B8E272
PrimaryGroupID: 510
RecordName: developers
RecordType: dsRecTypeStandard:Groups

```


Renaming group name and deleting the group

```
<dscf_cmd> DS Error: -14134 (eDSAttributeNotFound)
waseemakram@waseem ~ % dscl . -list /groups | grep developers

developers
waseemakram@waseem ~ % sudo dscl . -change /Groups/developers RecordName developers devteam

waseemakram@waseem ~ % dscl . -list /groups | grep devteam

devteam
[waseemakram@waseem ~ % dscl . -delete /groups/devteam

<main> delete status: eDSPermissionError
<dscf_cmd> DS Error: -14120 (eDSPermissionError)
[waseemakram@waseem ~ % sudo dscl . -delete /groups/devteam

waseemakram@waseem ~ % dscl . -list /groups | grep devteam

waseemakram@waseem ~ % █
```

Id : shoes information of user

```

waseemakram@waseem ~ % id waseem
uid=510(waseem) gid=20(staff) groups=20(staff),5010(devops),12(everyone),6010(admin.apple.access_ssh),400(com.apple.access_remote_ae),701(com.apple.sharepoint.admin)
waseemakram@waseem ~ % getent passwd waseem
sh: command not found: getent
waseemakram@waseem ~ % dscl . read /users/waseem

dsAttrTypeNative:_writers_passwd: waseem
dsAttrTypeNative:accountPolicyData:
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<dict>
  <key>creationTime</key>
  <real>1760089357.961612</real>
  <key>failedLoginCount</key>
  <integer>10</integer>
  <key>failedLoginTimestamp</key>
  <real>1761720499.5167201</real>
  <key>passwordLastSetTime</key>
  <real>1760089764.2840819</real>
</dict>
</plist>

dsAttrTypeNative:record_daemon_version: 9054000
appleMetaNodeLocation: /Local/Default
generatedUID: CAB257AE-B108-43DD-BF66-F45C624819B4
FSHomeDirectory: /Users/waseem
password: *****
primaryGroupID: 20
realName:
Waseem Akram
recordName: waseem
recordType: dsRecTypeStandard:Users
uniqueID: 510
userShell: /bin/zsh

```

W :shoes logged in users and details
 Who: shows currently logged in users
 Vipw :edits the passwd file

```

userShell: /bin/zsh
[waseemakram@waseem ~ % passwd waseem
Changing password for waseem.
[Old Password:
[waseemakram@waseem ~ % w
10:31 up 2 days, 10:05, 2 users, load averages: 1.05 1.22 1.40
USER      TTY      FROM      LOGIN@   IDLE WHAT
waseemakra console -        Sat00    2days -
waseemakra s000 -        Sat00    - w
[waseemakram@waseem ~ % who
waseemakram      console          1 Nov 00:27
waseemakram      ttys000          1 Nov 00:27
[waseemakram@waseem ~ % vipw
vipw: /etc/master.passwd: Permission denied
waseemakram@waseem ~ % █

```

```

waseemakram@waseem waseem % chgrp staff file2
waseemakram@waseem waseem % ls -l file2
-rwxr-xr-x 1 waseemakram staff 22 2 Nov 12:39 file2
waseemakram@waseem waseem % sudo chown root file2
Password:
waseemakram@waseem waseem % ls -l file2
-rwxr-xr-x 1 root staff 22 2 Nov 12:39 file2
waseemakram@waseem waseem % diff file2 file4
1d0
< hello this is waseem
waseemakram@waseem waseem % find / -name file4
find: /usr/sbin/authserver: Permission denied
find: /Library/Application Support/Apple/ParentalControls/Users: Permission denied
find: /Library/Application Support/Apple/AssetCache/Data: Permission denied

```

```

[waseemakram@waseem waseem % ls -ll
total 48
drwxr-xr-x 3 waseemakram staff 96 31 Oct 17:23 akram
-rw-r--r-- 1 waseemakram staff 0 22 Oct 14:55 akram.git
-rw-r--r-- 1 waseemakram staff 0 22 Oct 16:10 app
-rw-r--r-- 1 waseemakram staff 63 1 Nov 11:03 file1.bz2
-rw-r--r-- 1 waseemakram staff 116 31 Oct 17:11 file1.gz.xz
-rw-r--r-- 1 waseemakram staff 22 2 Nov 12:39 file2
-rw-r--r-- 1 waseemakram staff 0 2 Nov 12:39 file4
-rw-r--r-- 1 waseemakram staff 160 2 Nov 12:39 file4.zip
-rw-r--r-- 1 waseemakram staff 0 22 Oct 16:28 games
-rwxr-xr-x 1 waseemakram staff 1024 31 Oct 17:56 horizon
-rw-r--r-- 1 waseemakram staff 2560 31 Oct 17:58 horizon.tar
lrwxr-xr-x 1 waseemakram staff 5 31 Oct 17:18 soft.lnk -> file1
[waseemakram@waseem waseem % chmod 755 file2
[waseemakram@waseem waseem % ls -ll
total 48
drwxr-xr-x 3 waseemakram staff 96 31 Oct 17:23 akram
-rw-r--r-- 1 waseemakram staff 0 22 Oct 14:55 akram.git
-rw-r--r-- 1 waseemakram staff 0 22 Oct 16:10 app
-rw-r--r-- 1 waseemakram staff 63 1 Nov 11:03 file1.bz2
-rw-r--r-- 1 waseemakram staff 116 31 Oct 17:11 file1.gz.xz
-rwxr-xr-x 1 waseemakram staff 22 2 Nov 12:39 file2
-rw-r--r-- 1 waseemakram staff 0 2 Nov 12:39 file4
-rw-r--r-- 1 waseemakram staff 160 2 Nov 12:39 file4.zip
-rw-r--r-- 1 waseemakram staff 0 22 Oct 16:28 games
-rwxr-xr-x 1 waseemakram staff 1024 31 Oct 17:56 horizon
-rw-r--r-- 1 waseemakram staff 2560 31 Oct 17:58 horizon.tar
lrwxr-xr-x 1 waseemakram staff 5 31 Oct 17:18 soft.lnk -> file1
[waseemakram@waseem waseem % ls -ll | grep file2
-rwxr-xr-x 1 waseemakram staff 22 2 Nov 12:39 file2
[waseemakram@waseem waseem % chgrp file2
usage: chgrp [-fhvnx] [-R [-H | -L | -P]] group file ...
[waseemakram@waseem waseem % chgrp staff file2
[waseemakram@waseem waseem % ls -l file2
-rwxr-xr-x 1 waseemakram staff 22 2 Nov 12:39 file2

```

Chang

chgrp: change group of file

Chown: changes the owner of the file

Diff : differences the two files

Find : finds the file Umask sets the default permissions for newly created files and directories.

Lsof : list opens file

It

```
waseemakram@waseem waseem % umask 022

waseemakram@waseem waseem % touch testfile
waseemakram@waseem waseem % mkdir testdir
waseemakram@waseem waseem % ls -l
total 48
-rwxr-xr-x  3 waseemakram  staff    96 31 Oct 17:23 akram
-rw-r--r--  1 waseemakram  staff     0 22 Oct 14:55 akram.git
-rw-r--r--  1 waseemakram  staff     0 22 Oct 16:10 app
-rw-r--r--  1 waseemakram  staff    63  1 Nov 11:03 file1.bz2
-rw-r--r--  1 waseemakram  staff   116 31 Oct 17:11 file1.gz.xz
-rwxr-xr-x  1 root         staff    22  2 Nov 12:39 file2
-rw-r--r--  1 waseemakram  staff     0  2 Nov 12:39 file4
-rw-r--r--  1 waseemakram  staff   160  2 Nov 12:39 file4.zip
-rw-r--r--  1 waseemakram  staff     0 22 Oct 16:28 games
-rwxr-xr-x  1 waseemakram  staff  1024 31 Oct 17:56 horizon
-rw-r--r--  1 waseemakram  staff  2560 31 Oct 17:58 horizon.tar
-rwxr-xr-x  1 waseemakram  staff     5 31 Oct 17:18 soft.lnk -> file1
-rwxr-xr-x  2 waseemakram  staff    64  3 Nov 10:59 testdir
-rw-r--r--  1 waseemakram  staff     0  3 Nov 10:59 testfile
```

shows

which **files, ports, or network connections** are currently open — useful for debugging or finding what's using a file or port.

```
waseemakram@waseem waseem % lsof -i :8081

COMMAND  PID      USER  FD  TYPE             DEVICE  SIZE/OFF  NODE NAME
nginx    10197 waseemakram  6u  IPv4 0xb80a47776e77cb12  0t0  TCP *:sunproxyadmin (LISTEN)
nginx    10198 waseemakram  6u  IPv4 0xb80a47776e77cb12  0t0  TCP *:sunproxyadmin (LISTEN)
waseemakram@waseem waseem % lsof -i :8080

waseemakram@waseem waseem % lsof -i :82

waseemakram@waseem waseem % lsof -i :81

waseemakram@waseem waseem % mount
/dev/disk3s1s1 on / (apfs, sealed, local, read-only, journaled)
devfs on /dev (devfs, local, nobrowse)
/dev/disk3s6 on /System/Volumes/VM (apfs, local, noexec, journaled, noatime, nobrowse)
/dev/disk3s2 on /System/Volumes/Preboot (apfs, local, journaled, nobrowse)
/dev/disk3s4 on /System/Volumes/Update (apfs, local, journaled, nobrowse)
/dev/disk1s2 on /System/Volumes/xarts (apfs, local, noexec, journaled, noatime, nobrowse)
/dev/disk1s1 on /System/Volumes/iSCPreboot (apfs, local, journaled, nobrowse)
/dev/disk1s3 on /System/Volumes/Hardware (apfs, local, journaled, nobrowse)
/dev/disk3s5 on /System/Volumes/Data (apfs, local, journaled, nobrowse, protect, root data)
map auto_home on /System/Volumes/Data/home (autofs, automounted, nobrowse)
/Users/waseemakram/Downloads/Visual Studio Code.app on /private/var/folders/g7/ldjy8qpd1458ss352fg6tv4m0000gn/T/AppTransloc
/Users/waseemakram/Downloads/Sublime Text.app on /private/var/folders/g7/ldjy8qpd1458ss352fg6tv4m0000gn/T/AppTransloc
waseemakram@waseem waseem % findmnt
```

Systemctl start nginx starts the nginx server

Systemctl status nginx shoes status of the nginx server


```

[ec2-user@ip-172-31-7-102 ~]$ sudo yum install nginx
Last metadata expiration check: 0:01:27 ago on Mon Nov  3 06:19:49 2025.
Package nginx-1:1.28.0-1.amzn2023.0.2.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: inactive (dead)
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Mon 2025-11-03 06:21:47 UTC; 4s ago
     Process: 2517 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 2518 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 2519 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 2520 (nginx)
      Tasks: 3 (limit: 2248)
     Memory: 4.9M
        CPU: 54ms
    CGroup: /system.slice/nginx.service
            └─2520 "nginx: master process /usr/sbin/nginx"
              └─2521 "nginx: worker process"
                └─2522 "nginx: worker process"

```

Systemctl restart :restarts the server

Systemctl stop :stops the running server

Systemctl enable:a unit file to start automatically on system boot and

```

[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl stop httpd
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl stop nginx
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: inactive (dead)

Nov 03 06:21:47 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Starting nginx.service - Th
Nov 03 06:21:47 ip-172-31-7-102.eu-north-1.compute.internal nginx[2518]: nginx: the configuration f
Nov 03 06:21:47 ip-172-31-7-102.eu-north-1.compute.internal nginx[2518]: nginx: configuration file

```

systemctl disable :Disables a unit file from starting automatically on boot.

systemctl list-units :Lists all currently loaded unit files.

sudo systemctl set-default graphical.target:

means that the system will now **boot into GUI mode by default** — *but only if a graphical desktop environment (like GNOME or KDE) is installed.*

Systemctl edit nginx **Opens an editor to modify a unit file.**

```

[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl disable nginx
Removed "/etc/systemd/system/multi-user.target.wants/nginx.service".
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl list-units

```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
boot-efi.automount	loaded	active	running	boot-efi.automount
proc-sys-fs-binfmt_misc.automount	loaded	active	waiting	Arbitrary Executable File F
sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1-nvme0nlp1.device	loaded	active	plugged	Amazon Elastic Block Store
sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1-nvme0nlp127.device	loaded	active	plugged	Amazon Elastic Block Store
sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1-nvme0nlp128.device	loaded	active	plugged	Amazon Elastic Block Store
sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1.device	loaded	active	plugged	Amazon Elastic Block Store
sys-devices-pci0000:00-0000:00:05.0-net-ens5.device	loaded	active	plugged	Elastic Network Adapter (EN
sys-devices-platform-serial8250-tty-ttyS1.device	loaded	active	plugged	/sys/devices/platform/seria
sys-devices-platform-serial8250-tty-ttyS2.device	loaded	active	plugged	/sys/devices/platform/seria
sys-devices-platform-serial8250-tty-ttyS3.device	loaded	active	plugged	/sys/devices/platform/seria
sys-devices-pnp0-00:04-tty-ttyS0.device	loaded	active	plugged	/sys/devices/pnp0/00:04/tty
sys-module-configfs.device	loaded	active	plugged	/sys/module/configfs
sys-module-fuse.device	loaded	active	plugged	/sys/module/fuse
sys-subsystem-net-devices-ens5.device	loaded	active	plugged	Elastic Network Adapter (EN
-.mount	loaded	active	mounted	Root Mount
boot-efi.mount	loaded	active	mounted	/boot/efi
dev-hugepages.mount	loaded	active	mounted	Huge Pages File System
dev-mqueue.mount	loaded	active	mounted	POSIX Message Queue File Sys

```

[ec2-user@ip-172-31-29-104 ~]$ systemctl set-default graphical.target
Failed to set default target: Access denied
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl set-default graphical.target
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl get-default
graphical.target
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl cat nginx
# /usr/lib/systemd/system/nginx.service
[Unit]
Description=The nginx HTTP and reverse proxy server
After=network-online.target remote-fs.target nss-lookup.target
Wants=network-online.target

[Service]
Type=forking
PIDFile=/run/nginx.pid
# Nginx will fail to start if /run/nginx.pid already exists but has the wrong
# SELinux context. This might happen when running `nginx -t` from the cmdline.
# https://bugzilla.redhat.com/show_bug.cgi?id=1268621
ExecStartPre=/usr/bin/rm -f /run/nginx.pid
ExecStartPre=/usr/sbin/nginx -t
ExecStart=/usr/sbin/nginx
ExecReload=/usr/sbin/nginx -s reload
KillSignal=SIGQUIT
TimeoutStopSec=5
KillMode=mixed
PrivateTmp=true

[Install]
WantedBy=multi-user.target

```

systemctl daemon-reload:

Instructs the systemctl main process to reload its configuration.

```

GNU nano 8.3 /etc/systemd/system/nginx.service.d/override.conf
### Editing /etc/systemd/system/nginx.service.d/override.conf
### Anything between here and the comment below will become the new contents of the file

### Lines below this comment will be discarded

### /usr/lib/systemd/system/nginx.service
# [Unit]
# Description=The nginx HTTP and reverse proxy server
# After=network-online.target remote-fs.target nss-lookup.target
# Wants=network-online.target
#
# [Service]
# Type=forking
# PIDFile=/run/nginx.pid
# # Nginx will fail to start if /run/nginx.pid already exists but has the wrong
# # SELinux context. This might happen when running `nginx -t` from the cmdline.
# # https://bugzilla.redhat.com/show_bug.cgi?id=1268621
# ExecStartPre=/usr/bin/rm -f /run/nginx.pid
# ExecStartPre=/usr/sbin/nginx -t
# ExecStart=/usr/sbin/nginx
# ExecReload=/usr/sbin/nginx -s reload
# KillSignal=SIGQUIT
# TimeoutStopSec=5
# KillMode=mixed
# PrivateTmp=true
#
# [Install]
# WantedBy=multi-user.target

```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
 ^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify ^_/ Go To Line M-E Redo M-G Copy ^B Where Was

Systemctl daemon-reload :instructs the systemctl main process to reload its configuration
 Systemctl isolate Switches the system to a different target.
 Systemctl list-dependencies Lists all dependencies of a specific target.

```
complete.
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-29-104 ~]$ systemctl edit nginx
Failed to open "/etc/systemd/system/nginx.service.d/.#override.conf225649e458eld1bd": Permission denied
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl edit nginx
Editing "/etc/systemd/system/nginx.service.d/override.conf" canceled: temporary file is empty.
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl daemon-reload
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl isolate graphical.target
[ec2-user@ip-172-31-29-104 ~]$ sudo systemctl list-dependencies graphical.target
graphical.target
├─display-manager.service
├─systemd-update-utmp-runlevel.service
├─multi-user.target
│   ├── amazon-ssm-agent.service
│   ├── atd.service
│   ├── auditd.service
│   ├── chronyd.service
│   ├── hibinit-agent.service
│   ├── irqbalance.service
│   ├── libstoragemgmt.service
│   ├── sshd.service
│   ├── sssd.service
│   ├── sysstat.service
│   └─systemd-ask-password-wall.path
```

Rmp : shows us the package details info

Yum: is a software package manager used in Red Hat-based environments for installing, updating, and managing packages

```
[ec2-user@ip-172-31-29-104 ~]$ rpm -q nginx
nginx-1.28.0-1.amzn2023.0.2.x86_64
[ec2-user@ip-172-31-29-104 ~]$ sudo yum install http
Last metadata expiration check: 0:39:15 ago on Mon Nov  3 06:33:20 2025.
No match for argument: http
Error: Unable to find a match: http
[ec2-user@ip-172-31-29-104 ~]$ sudo yum install httpd
Last metadata expiration check: 0:39:23 ago on Mon Nov  3 06:33:20 2025.
Dependencies resolved.
=====
Package                                     Architecture
=====
Installing:
  httpd                                     x86_64
Installing dependencies:
  apr                                       x86_64
  apr-util                                 x86_64
```

Dnf :Similar to yum, dnf is a modern package manager for Red Hat-based systems, offering improved performance and package management capabilities.

```
[ec2-user@ip-172-31-29-104 ~]$ dnf update
Error: This command has to be run with superuser privileges (under the root user on most systems).
[ec2-user@ip-172-31-29-104 ~]$ sudo dnf update
Last metadata expiration check: 0:42:06 ago on Mon Nov 3 06:33:20 2025.
Dependencies resolved.
Nothing to do.
Complete!
```

Apt: apt is a command-line tool used in Ubuntu and other Debian-based systems for handling packages, including installation, updating, and removal. Apt cache is used for querying package information in Debian-based systems. It's especially useful for searching for packages containing specific files.

```
ubuntu@ip-172-31-17-3:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-b
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libapru
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 2086 kB of archives.
After this operation, 8090 kB of additional disk space will be used
Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/main am
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/main am
Get:4 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/main am
```

- `sudo vim /etc/ssh/sshd_config`

Will open the editor for changing sshd configuration port

Update SSH Port to 2022

```
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (
# configuration must be re-generated after
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 2022
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none
```


- **sudo systemctl restart sshd:**
 - Restart the SSH daemon to apply the new configuration.
- Sudo systemctl status sshd will show the status of sshd

```

failed to start sshd.service: Unit sshd.service not found.
ubuntu@ip-172-31-17-3:~$ sudo systemctl restart sshd
ubuntu@ip-172-31-17-3:~$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/ssh.service.d
            └─ec2-instance-connect.conf
   Active: active (running) since Mon 2025-11-03 07:38:26 UTC; 28s ago
 TriggeredBy: ● ssh.socket
     Docs: man:sshd(8).
           man:sshd_config(5).
   Process: 2472 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 2474 (sshd)
    Tasks: 1 (limit: 1008)
   Memory: 1.2M (peak: 1.5M)
      CPU: 20ms
   CGroup: /system.slice/ssh.service
           └─2474 "sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ec2-instance-

Nov 03 07:38:26 ip-172-31-17-3 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server.
Nov 03 07:38:26 ip-172-31-17-3 sshd[2474]: Server listening on 0.0.0.0 port 22.
Nov 03 07:38:26 ip-172-31-17-3 sshd[2474]: Server listening on :: port 22.
Nov 03 07:38:26 ip-172-31-17-3 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
lines 1-20/20 (END)
[1]+  Stopped                  sudo systemctl status sshd
ubuntu@ip-172-31-17-3:~$ sudo grep Port /etc/ssh/sshd_config
# configuration must be re-generated after changing Port, AddressFamily, or
#Port 2022
#GatewayPorts no
ubuntu@ip-172-31-17-3:~$ sudo grep Port /etc/ssh/sshd_config

```

Chronyc: interfaces with the `chronyd` time-synchronizing service, providing detailed information

```
[ec2-user@ip-172-31-7-102 ~]$ chronyc tracking
Reference ID      : A9FEA97B (169.254.169.123)
Stratum          : 4
Ref time (UTC)   : Mon Nov 03 09:38:01 2025
System time      : 0.000003174 seconds slow of NTP time
Last offset      : +0.000000319 seconds
RMS offset       : 0.000037402 seconds
Frequency        : 0.508 ppm fast
Residual freq    : +0.004 ppm
Skew             : 0.636 ppm
Root delay       : 0.000256788 seconds
Root dispersion  : 0.000228487 seconds
Update interval  : 16.6 seconds
Leap status      : Normal
[ec2-user@ip-172-31-7-102 ~]$
```

Manage time commands

Date: displays the current time and date

Hwclock: Used for managing and viewing the hardware clock. It can also be used to synchronize the hardware clock with the system clock.

```
hwclock: cannot access the hardware clock via any known method.
[ec2-user@ip-172-31-7-102 ~]$ sudo hwclock --verbose
hwclock from util-linux 2.37.4
System Time: 1762162623.965658
Trying to open: /dev/rtc0
Using the rtc interface to the clock.
Last drift adjustment done at 0 seconds after 1969
Last calibration done at 0 seconds after 1969
Hardware clock is on UTC time
Assuming hardware clock is kept in UTC time.
Waiting for clock tick...
...got clock tick
Time read from Hardware Clock: 2025/11/03 09:37:05
Hw clock time : 2025/11/03 09:37:05 = 1762162625 seconds since 1969
Time since last adjustment is 1762162625 seconds
Calculated Hardware Clock drift is 0.000000 seconds
2025-11-03 09:37:03.462860+00:00
[ec2-user@ip-172-31-7-102 ~]$ timedatectl status
                Local time: Mon 2025-11-03 09:37:39 UTC
                Universal time: Mon 2025-11-03 09:37:39 UTC
                   RTC time: Mon 2025-11-03 09:37:39
                   Time zone: n/a (UTC, +0000)
System clock synchronized: yes
                   NTP service: active
                   RTC in local TZ: no
```

Process Management Commands

```
[ec2-user@ip-172-31-7-102 ~]$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.8 172480 17432 ?        Ss   09:35   0:00 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root           2  0.0  0.0      0     0 ?        S    09:35   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   09:35   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   09:35   0:00 [rcu_par_gp]
root           5  0.0  0.0      0     0 ?        I<   09:35   0:00 [slub_flushwq]
root           6  0.0  0.0      0     0 ?        I<   09:35   0:00 [netns]
root           8  0.0  0.0      0     0 ?        I<   09:35   0:00 [kworker/0:0H-events_highpri]
root           9  0.0  0.0      0     0 ?        I    09:35   0:00 [kworker/u4:0-flush-259:0]
root          10  0.0  0.0      0     0 ?        I<   09:35   0:00 [mm_percpu_wq]
root          11  0.0  0.0      0     0 ?        I    09:35   0:00 [rcu_tasks_kthread]
root          12  0.0  0.0      0     0 ?        I    09:35   0:00 [rcu_tasks_rude_kthread]
root          13  0.0  0.0      0     0 ?        I    09:35   0:00 [rcu_tasks_trace_kthread]
root          14  0.0  0.0      0     0 ?        S    09:35   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    09:35   0:00 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    09:35   0:00 [migration/0]
root          17  0.0  0.0      0     0 ?        I    09:35   0:00 [kworker/0:1-events]
root          18  0.0  0.0      0     0 ?        S    09:35   0:00 [cpuhp/0]
root          19  0.0  0.0      0     0 ?        S    09:35   0:00 [cpuhp/1]
root          20  0.0  0.0      0     0 ?        S    09:35   0:00 [migration/1]
root          21  0.0  0.0      0     0 ?        S    09:35   0:00 [ksoftirqd/1]
root          23  0.0  0.0      0     0 ?        I<   09:35   0:00 [kworker/1:0H-events_highpri]
root          26  0.0  0.0      0     0 ?        S    09:35   0:00 [kdevtmpfs]
root          27  0.0  0.0      0     0 ?        I<   09:35   0:00 [inet_frag_wq]
root          28  0.0  0.0      0     0 ?        S    09:35   0:00 [kauditd]
root          29  0.0  0.0      0     0 ?        S    09:35   0:00 [khungtaskd]
```

Top :Monitors all processes and the general health of the system from a performance perspective.

Ps: process status

Ex: ps aux :Shows all processes and their properties; useful for monitoring system processes.

Jobs:list **background and suspended processes** started in your current terminal session.

Fg:Brings a job to the foreground, where %1 is the job number

Bg: Sends a job to run in the background.

MiB Swap:		0.0 total,	0.0 free,	0.0 used.	1106.6 avail	Mem					
PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
826	root	20	0	0	0	0	I	0.3	0.0	0:00.03	kworker/1:3-mm_percpu_wq
1454	jenkins	20	0	3344728	340728	26832	S	0.3	17.5	0:25.84	java
1486	root	20	0	3883928	137924	22556	S	0.3	7.1	0:05.55	java
2704	ec2-user	20	0	224024	3344	2784	R	0.3	0.2	0:00.02	top
1	root	20	0	172480	17432	10784	S	0.0	0.9	0:00.96	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	slub_flushwq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	netns
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
9	root	20	0	0	0	0	I	0.0	0.0	0:00.03	kworker/u4:0-flush-259:0
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
11	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
14	root	20	0	0	0	0	S	0.0	0.0	0:00.03	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.06	rcu_preempt
16	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
17	root	20	0	0	0	0	I	0.0	0.0	0:00.04	kworker/0:1-events
18	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
20	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/1
21	root	20	0	0	0	0	S	0.0	0.0	0:00.03	ksoftirqd/1
23	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_highpri
26	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
27	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	inet_frag_wq
28	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
29	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungtaskd
30	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
32	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	writeback
33	root	20	0	0	0	0	S	0.0	0.0	0:00.01	kcompactd0
34	root	39	19	0	0	0	S	0.0	0.0	0:00.00	khugepaged
35	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	cryptd
36	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kintegrityd

```

[waseemakram@waseem ~ % jobs
[1]      suspended  grep $1 file1
[2]      suspended  find / -mtime -1
[3]      suspended  grep -v waseemz
[4]      suspended  grep passwd
[5]      suspended  unzip file4
[6]      suspended  less david
[7]      suspended  tee
[8]      suspended  find / -name file4
[9]      suspended  find / -name file2
[10]     suspended  sudo find / -name file2
[11]     - suspended  sudo find / -name nginx.conf
[12]     + suspended  ping google.com
[waseemakram@waseem ~ % fg %5

[5]      continued  unzip file4
(pwd : ~/waseem)
NULL
(EOF or read error, treating as "[N]one" ...)
(pwd now: ~)
waseemakram@waseem ~ % file file4

file4: cannot open `file4' (No such file or directory)
[waseemakram@waseem ~ % bg %5

bg: %5: no such job
[waseemakram@waseem ~ % jobs
[1]      suspended  grep $1 file1
[2]      suspended  find / -mtime -1
[3]      suspended  grep -v waseemz
[4]      suspended  grep passwd
[6]      suspended  less david
[7]      suspended  tee
[8]      suspended  find / -name file4
[9]      suspended  find / -name file2
[10]     suspended  sudo find / -name file2
[11]     - suspended  sudo find / -name nginx.conf
[12]     + suspended  ping google.com
[waseemakram@waseem ~ % kill %1 %2
[1]      terminated  grep $1 file1
[2]      terminated  find / -mtime -1
waseemakram@waseem ~ % █

```

Kill: for terminating a process

Nice: The **nice** command is actually used to **set the priority** of a process when you start it.

```

11010 -o ping
waseemakram@waseem ~ % nice -n 10 sleep 60 &
ps -o pid,ni,comm | grep sleep

[1] 15075
15075 15 sleep

```

Renice: changes the priority of an already running process


```
waseemakram@waseem ~ % ps -o pid,ni,comm | grep ping
11810 0 ping
waseemakram@waseem ~ % sudo renice -n 10 -p 11810
Password:
waseemakram@waseem ~ % ps -o pid,ni,comm | grep 11810
11810 10 ping
waseemakram@waseem ~ %
```

Scheduling Tasks Commands

At: Allows you to schedule a job to run at a specific time.

```
waseemakram@waseem ~ % at 3:57 PM

echo "Hello Waseem, this ran at 3:59!" > ~/Desktop/at_test.txt
^[[A^[[D

job 1 at Mon Nov 3 15:57:00 2025
waseemakram@waseem ~ % atq

1      Mon Nov 3 15:57:00 2025
waseemakram@waseem ~ % cat ~/Desktop/at_test.txt
```

Atq is used to **list all the pending jobs** that have been scheduled using the at command.
is used to **list**

```
Hello Waseem, at test successful!
[waseemakram@waseem ~ % atq
1      Mon Nov 3 15:57:00 2025
[waseemakram@waseem ~ % atrm 1
[waseemakram@waseem ~ % atq
waseemakram@waseem ~ %
```

atrm:removes the job

Crontab -e:Opens the cron configuration for editing, allowing you to schedule jobs to run periodically.

- opens the cron configuration file in a text editor.

```
waseemakram@waseem ~ % crontab -e

crontab: installing new crontab
waseemakram@waseem ~ % cat ~/cron_test.txt

Testing cron for Waseem!
waseemakram@waseem ~ %
```

- `systemctl list-timers` shows all active timers.

```
[ec2-user@ip-172-31-7-102 ~]$ systemctl list-timers
NEXT LEFT LAST PASSED UNIT ACTIVATION
Mon 2025-11-03 10:51:04 UTC 2s left - refresh-policy-routes@ens5.timer refresh-policy-routes@ens5.timer
Mon 2025-11-03 11:00:00 UTC 8min left - sysstat-collect.timer sysstat-collect.timer
Mon 2025-11-03 11:05:28 UTC 14min left - systemd-tmpfiles-clean.timer systemd-tmpfiles-clean.timer
Mon 2025-11-03 12:08:57 UTC 1h 17min left Wed 2025-10-29 14:08:17 UTC 4 days ago fstrim.timer fstrim.timer
Mon 2025-11-03 23:32:37 UTC 12h left Fri 2025-10-31 01:26:18 UTC 3 days ago update-motd.timer update-motd.timer
Tue 2025-11-04 00:00:00 UTC 13h left Mon 2025-11-03 06:19:40 UTC 4h 31min ago logrotate.timer logrotate.timer
Tue 2025-11-04 00:07:00 UTC 13h left - sysstat-summary.timer sysstat-summary.timer

7 timers listed.
Pass --all to see loaded but inactive timers, too.
[ec2-user@ip-172-31-7-102 ~]$
```

Reading Log Files Commands

Journalctl: This command is used to check the contents of the systemd journal, which is a centralized logging system for Linux.

- This example shows how to view logs for the Nginx service using `journalctl`.

```
Pass --all to see loaded but inactive timers, too.
[ec2-user@ip-172-31-7-102 ~]$ journalctl -u nginx.service
Oct 19 08:14:07 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Starting nginx.service - The nginx M
Oct 19 08:14:07 ip-172-31-7-102.eu-north-1.compute.internal nginx[5397]: nginx: the configuration file /etc
Oct 19 08:14:07 ip-172-31-7-102.eu-north-1.compute.internal nginx[5397]: nginx: configuration file /etc/ngin
Oct 19 08:14:07 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Started nginx.service - The nginx H
Oct 19 08:43:43 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Stopping nginx.service - The nginx M
Oct 19 08:43:43 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: nginx.service: Deactivated successf
Oct 19 08:43:43 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Stopped nginx.service - The nginx H
-- Boot 38dfa40bdbcc4e0fba95a5f15c8ae546 --
Oct 23 10:38:57 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Starting nginx.service - The nginx M
Oct 23 10:38:57 ip-172-31-7-102.eu-north-1.compute.internal nginx[2271]: nginx: the configuration file /etc
Oct 23 10:38:57 ip-172-31-7-102.eu-north-1.compute.internal nginx[2271]: nginx: configuration file /etc/ngin
Oct 23 10:38:57 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Started nginx.service - The nginx H
Oct 23 11:18:46 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Stopping nginx.service - The nginx M
Oct 23 11:18:46 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: nginx.service: Deactivated successf
Oct 23 11:18:46 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Stopped nginx.service - The nginx H
-- Boot 4cb9dd01f98d47d0ac04d8eef1524d65 --
Oct 29 13:05:27 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Starting nginx.service - The nginx M
Oct 29 13:05:27 ip-172-31-7-102.eu-north-1.compute.internal nginx[3448]: nginx: the configuration file /etc
Oct 29 13:05:27 ip-172-31-7-102.eu-north-1.compute.internal nginx[3448]: nginx: configuration file /etc/ngin
Oct 29 13:05:27 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Started nginx.service - The nginx H
Oct 29 13:13:35 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Stopping nginx.service - The nginx M
Oct 29 13:13:35 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: nginx.service: Deactivated successf
Oct 29 13:13:35 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Stopped nginx.service - The nginx H
```

Head

```
waseemakram@waseem ~ % head -n 20 /var/log/system.log

Nov  3 00:08:38 waseem syslogd[126]: ASL Sender Statistics
Nov  3 00:20:59 waseem syslogd[126]: ASL Sender Statistics
Nov  3 00:30:02 waseem syslogd[126]: Configuration Notice:
    ASL Module "com.apple.cdscheduler" claims selected messages.
    Those messages may not appear in standard system log files or in the ASL database.
Nov  3 00:30:02 waseem syslogd[126]: Configuration Notice:
    ASL Module "com.apple.install" claims selected messages.
    Those messages may not appear in standard system log files or in the ASL database.
Nov  3 00:30:02 waseem syslogd[126]: Configuration Notice:
    ASL Module "com.apple.authd" sharing output destination "/var/log/asl" with ASL Module "com.apple.asl".
    Output parameters from ASL Module "com.apple.asl" override any specified in ASL Module "com.apple.authd".
Nov  3 00:30:02 waseem syslogd[126]: Configuration Notice:
    ASL Module "com.apple.authd" sharing output destination "/var/log/system.log" with ASL Module "com.apple.asl".
    Output parameters from ASL Module "com.apple.asl" override any specified in ASL Module "com.apple.authd".
Nov  3 00:30:02 waseem syslogd[126]: Configuration Notice:
    ASL Module "com.apple.authd" claims selected messages.
    Those messages may not appear in standard system log files or in the ASL database.
Nov  3 00:30:02 waseem syslogd[126]: Configuration Notice:
    ASL Module "com.apple.eventmonitor" claims selected messages.
    Those messages may not appear in standard system log files or in the ASL database.
waseemakram@waseem ~ %
```

Tail command is used to view the contents of log files, typically the last few lines.

```
Those messages may not appear in standard system log files or in the A
waseemakram@waseem ~ % tail -n 50 /var/log/system.log
Nov  3 05:44:04 waseem syslogd[126]: ASL Sender Statistics
Nov  3 05:59:33 waseem syslogd[126]: ASL Sender Statistics
Nov  3 06:14:56 waseem syslogd[126]: ASL Sender Statistics
Nov  3 06:30:26 waseem syslogd[126]: ASL Sender Statistics
Nov  3 06:48:14 waseem syslogd[126]: ASL Sender Statistics
Nov  3 07:03:39 waseem syslogd[126]: ASL Sender Statistics
Nov  3 07:19:06 waseem syslogd[126]: ASL Sender Statistics
Nov  3 07:34:30 waseem syslogd[126]: ASL Sender Statistics
Nov  3 07:50:00 waseem syslogd[126]: ASL Sender Statistics
Nov  3 08:05:23 waseem syslogd[126]: ASL Sender Statistics
Nov  3 08:20:48 waseem syslogd[126]: ASL Sender Statistics
Nov  3 08:34:06 waseem syslogd[126]: ASL Sender Statistics
Nov  3 08:49:30 waseem syslogd[126]: ASL Sender Statistics
Nov  3 09:05:00 waseem syslogd[126]: ASL Sender Statistics
Nov  3 09:15:34 waseem syslogd[126]: ASL Sender Statistics
Nov  3 09:32:51 waseem syslogd[126]: ASL Sender Statistics
Nov  3 09:51:35 waseem syslogd[126]: ASL Sender Statistics
Nov  3 10:03:55 waseem syslogd[126]: ASL Sender Statistics
Nov  3 10:14:14 waseem syslogd[126]: ASL Sender Statistics
Nov  3 10:25:13 waseem syslogd[126]: ASL Sender Statistics
Nov  3 10:36:14 waseem syslogd[126]: ASL Sender Statistics
Nov  3 10:49:13 waseem syslogd[126]: ASL Sender Statistics
Nov  3 11:03:28 waseem syslogd[126]: ASL Sender Statistics
Nov  3 11:16:34 waseem syslogd[126]: ASL Sender Statistics
Nov  3 11:27:25 waseem syslogd[126]: ASL Sender Statistics
Nov  3 11:45:31 waseem syslogd[126]: ASL Sender Statistics
Nov  3 12:04:48 waseem syslogd[126]: ASL Sender Statistics
Nov  3 12:16:10 waseem syslogd[126]: ASL Sender Statistics
Nov  3 12:26:11 waseem syslogd[126]: ASL Sender Statistics
Nov  3 12:36:18 waseem syslogd[126]: ASL Sender Statistics
Nov  3 12:46:43 waseem syslogd[126]: ASL Sender Statistics
Nov  3 12:57:11 waseem syslogd[126]: ASL Sender Statistics
Nov  3 13:08:01 waseem syslogd[126]: ASL Sender Statistics
Nov  3 13:18:09 waseem syslogd[126]: ASL Sender Statistics
Nov  3 13:33:09 waseem syslogd[126]: ASL Sender Statistics
Nov  3 13:51:31 waseem syslogd[126]: ASL Sender Statistics
Nov  3 14:07:53 waseem syslogd[126]: ASL Sender Statistics
Nov  3 14:23:47 waseem syslogd[126]: ASL Sender Statistics
Nov  3 14:40:56 waseem syslogd[126]: ASL Sender Statistics
Nov  3 14:58:26 waseem syslogd[126]: ASL Sender Statistics
Nov  3 15:13:26 waseem syslogd[126]: ASL Sender Statistics
Nov  3 15:27:21 waseem syslogd[126]: ASL Sender Statistics
Nov  3 15:40:12 waseem syslogd[126]: ASL Sender Statistics
Nov  3 15:50:44 waseem syslogd[126]: ASL Sender Statistics
Nov  3 16:04:28 waseem syslogd[126]: ASL Sender Statistics
Nov  3 16:15:09 waseem syslogd[126]: ASL Sender Statistics
Nov  3 16:27:59 waseem syslogd[126]: ASL Sender Statistics
Nov  3 16:38:09 waseem syslogd[126]: ASL Sender Statistics
Nov  3 16:48:54 waseem syslogd[126]: ASL Sender Statistics
Nov  3 17:00:37 waseem syslogd[126]: ASL Sender Statistics
waseemakram@waseem ~ %
```

Logger: The `logger` command is a convenient tool for writing messages to the system log.


```
Nov 3 17:00:37 waseem syslogd[126]: ASL Sender Statistics
waseemakram@waseem ~ % less /var/log/system.log

waseemakram@waseem ~ % logger "This is a test log message"

waseemakram@waseem ~ % tail -n 10 /var/log/system.log

Nov 3 15:27:21 waseem syslogd[126]: ASL Sender Statistics
Nov 3 15:40:12 waseem syslogd[126]: ASL Sender Statistics
Nov 3 15:50:44 waseem syslogd[126]: ASL Sender Statistics
Nov 3 16:04:28 waseem syslogd[126]: ASL Sender Statistics
Nov 3 16:15:09 waseem syslogd[126]: ASL Sender Statistics
Nov 3 16:27:59 waseem syslogd[126]: ASL Sender Statistics
Nov 3 16:38:09 waseem syslogd[126]: ASL Sender Statistics
Nov 3 16:48:54 waseem syslogd[126]: ASL Sender Statistics
Nov 3 17:00:37 waseem syslogd[126]: ASL Sender Statistics
Nov 3 17:11:53 waseem syslogd[126]: ASL Sender Statistics
waseemakram@waseem ~ % logger "System backup completed"

waseemakram@waseem ~ % log show --predicate 'eventMessage contains "System backup completed"' --last 1h

Filtering the log data using "composedMessage CONTAINS "System backup completed""
Skipping info and debug messages, pass --info and/or --debug to include.
Timestamp          Thread      Type      Activity          PID    TTL
2025-11-03 17:13:41.794789+0530 0x998ab    Default    0x0              16960  0    logger: System backup completed
-----
Log      - Default:      1, Info:      0, Debug:      0, Error:      0, Fault:      0
Activity - Create:      0, Transition: 0, Actions:    0
```


